

A Two-Stage Prediction + Detection Framework for Real-Time Epileptic Seizure Monitoring

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Abstract—Wearable intelligent devices are essential for the health monitoring of epilepsy patients in nonhospital environments. The existing seizure detection algorithms for wearable health monitoring devices cannot achieve high sensitivity, short detection latency, low false alarm rate (FAR), as well as lightweight computing simultaneously. In this article, we propose a two-stage prediction + detection framework, PDNet, for real-time epileptic seizure monitoring. The proposed two-stage PDNet framework consists of a lightweight seizure predictor and a highly accurate seizure detector. Only when the first-stage seizure predictor forecasts an impending seizure, the second-stage seizure detector is activated to precisely and rapidly classify the seizure states, thereby significantly reducing the amount of computations. A semisupervised learning strategy is employed to enhance the decision boundary of the seizure predictor, which is used for electroencephalogram (EEG) preprocessing instead of prediction only. Soft labels are adopted to enable the seizure detector to precisely distinguish the seizure states. The proposed PDNet is evaluated using the CHB-MIT scalp EEG database. The proposed PDNet achieves 99.0% sensitivity, 0.43/h FAR, and 3.52-s detection latency with 3.02 M multiply–accumulate (MAC) operations, which are competitive compared to the state-of-the-art in terms of sensitivity, detection latency, FAR, and computation complexity. Furthermore, fine-grained information such as the occurrence process of seizures demonstrated by soft labels can help the caregivers or clinicians to come up with targeted healthcare and clinical treatments.

Index Terms—Electroencephalogram (EEG), epilepsy, seizure detection, semisupervised learning, soft label, two-stage framework.

I. INTRODUCTION

EPILEPSY is one of the most common neurological disorders, which is characterized by transient abnormal

disturbance of brain neurons. Epilepsy seizures affect approximately 1% of human beings worldwide [1], [2]. As an unpredictable and chronic disease, epileptic seizures can lead to unconsciousness at any time, which puts the patients in disastrous situations.

Electroencephalogram (EEG) signals are widely employed for the monitoring of epileptic seizures, during which spikes are frequently captured in EEG signals. In general, the EEG signals of epilepsy patients can be categorized into four states: interictal (between seizures), preictal (before the seizure), ictal (during the seizure), and postictal (after the seizure) [3].

Continuous monitoring of EEG signals is necessary for the healthcare of epilepsy patients in nonhospital environments, where EEG analysis algorithms could be embedded in wearable devices for early warning or close-loop neurostimulation. EEG monitoring for epilepsy patients is either for seizure prediction or seizure detection. Seizure prediction forecasts the onset of epileptic seizures by identifying the preictal state of EEG recordings. Alternatively, seizure detection captures epileptic seizures by identifying the ictal state of EEG recordings [4].

Reliable seizure prediction algorithms can help healthcare professionals or patients forecast the next seizure so that measures can be taken in advance to minimize the damage caused by seizures [5]. However, no universal predictive characteristic or preictal biomarker can forecast the exact time of the incoming seizure [6]. Despite the promising results [e.g., high sensitivity and low false alarm rate (FAR)] reported by the previously published seizure prediction methods, a lone seizure predictor can only provide a probability that a seizure may occur a few minutes or even hours later. However, the exact time of occurrence cannot be predicted, which makes the prediction results less meaningful.

Different from seizure prediction, seizure detection algorithms can only monitor ongoing seizures. Automatic seizure event detection algorithms [7] can be used for long-term EEG visual inspection or treatment monitoring, which may not require a high sensitivity or a low FAR. To target more challenging tasks, automatic seizure detection algorithms are developed to detect the onset and termination of seizures [8], which are used for alarming or close-loop neurostimulation. The electrophysiological changes typically occur before the clinical onset of seizures. Accurately and rapidly detecting the onset and termination of seizures is the key to developing seizure-alarming/closed-loop EEG systems. Seizure detectors

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with high sensitivity, short detection latency, and low FAR are essential for such applications. If these three requirements are met simultaneously, a seizure detector is sufficient for real-time EEG monitoring while seizure predictors need not be considered. Unfortunately, none of the existing seizure detectors can fully satisfy these requirements.

To achieve high sensitivity, short detection latency, and low FAR simultaneously for ultralow-power applications, there are three challenges for seizure detectors to tackle.

- 1) *Postprocessing*: Postprocessing (e.g., majority voting [46]) is typically employed by a number of seizure detectors (or even seizure predictors) to reduce FAR. However, postprocessing easily introduces large latency (even >30 s in [46]). The conflict between the detection latency and FAR cannot be solved by postprocessing.
- 2) *Hard Label*: Most of the existing seizure detectors adopt hard labels (HLs) to represent “seizure” and “nonseizure” states. However, part of the information contained in the four different seizure states (i.e., interictal, preictal, ictal, and postictal states) is lost. Distinguishing the four states can help the experts, patients, and caregivers better understand the process of a seizure. The extracted information would be valuable for healthcare and potentially helpful for enhancing the detection capability of the seizure detector.
- 3) *How to Achieve “Lightweight”?*: To be embedded in wearable devices, seizure prediction/detection algorithms have to be lightweight, while high classification performance has to be maintained simultaneously. The proportion of the length of seizure is typically less than 2% for epilepsy patients [9]. Some works adopt multistage classification schemes to reduce the overall computing complexity [11], [12], [13], [14]. Only when the first-stage lightweight coarse classifier detects “suspected seizure,” the following-stage fine classifier(s) would be activated to provide a final decision. The multistage classification schemes can significantly reduce the overall computational complexity as the fine classifiers only work for a short period. However, the cascaded coarse and fine classifiers introduce long detection latency. Furthermore, the overall performance of the multistage classification methods depends heavily on the performance of the first-stage lightweight classifier, which is difficult to achieve high sensitivity and high specificity simultaneously.

In this article, we propose a two-stage prediction + detection neural network model PDNet for real-time epileptic seizure monitoring, to achieve high detection accuracy, low FAR, short detection latency, and lightweight computing simultaneously. The proposed two-stage PDNet model consists of a lightweight seizure predictor and a highly accurate seizure detector. Only when the first-stage seizure predictor forecasts an impending seizure, the second-stage seizure detector is activated to precisely and rapidly classify the seizure states. The main contributions of this work are summarized as follows.

- 1) A lightweight predictor with temporal ensembling in the proposed model is used for preprocessing instead of prediction only, to achieve low FAR and short detection

latency, thereby avoiding the pitfalls in the traditional postprocessing methods.

- 2) We adopt soft labels [circular smooth labels (CSLs)] to represent the seizure states. Combined with the Gaussian focal loss, soft labels can facilitate the proposed seizure monitor to precisely recognize the process of a seizure.
- 3) By only activating the highly accurate seizure detector in the proposed model occasionally, the overall computing complexity of the proposed model is maintained at a minimal level during the process of monitoring.

II. RELATED WORKS

EEG-based epileptic seizure monitoring can be divided into two categories: seizure prediction and seizure detection. In this section, the merits and drawbacks of the existing methods for EEG-based seizure prediction and detection are discussed.

A. Epileptic Seizure Prediction

Previously, EEG-based seizure prediction algorithms relied on indicators for impending seizures. Seizures are predicted to come when the kinetic indicators are above or below a predefined threshold [49]. However, no universal predictive characteristic for seizures has been identified. Instead of searching universal indicators, some researchers perform seizure prediction by training a machine learning classifier to distinguish EEG signals between preictal and interictal states. Parvez and Paul [15] use the phase correlation to explore the spatiotemporal relationship of EEG signals in the preictal and interictal states and use SVM as a classifier to perform seizure prediction. Shiao et al. [16] extract the power spectrum and cross-channel correlation of EEG signals as features and also use SVM as a classifier.

Recently, deep learning models have been widely used in seizure prediction, which greatly improves the classification performance compared to conventional machine learning methods. Some studies combine feature extraction and deep learning models. Li et al. [35] use short-time Fourier transform (STFT) to extract time-frequency features and use convolutional neural network (CNN) as the classifier. Ozcan and Erturk [36] construct a 3-D representation of features and adopt 3D-CNN as a classifier for seizure prediction. In [39], a graph convolutional network (GCN) is used to capture the spatiotemporal domain information from EEG signals for seizure prediction. Some other works use deep learning models to automatically learn features from the raw EEG data. Li et al. [37], [38] use CNN to extract features from EEG signals for seizure prediction. Daoud and Bayoumi [17] use CNN to extract features from EEG signals and use recurrent neural network (RNN) as a classifier. Currently, seizure prediction methods are usually used for early warning. However, the exact time of seizure onset cannot be predicted. The uncertainty increases the tension of patients and leads to alarm fatigue, which makes the prediction results less meaningful.

Another issue with the current seizure prediction methods is the division of preictal and interictal states. In EEG recordings, the duration of preictal and interictal states is not strictly defined, and the boundary between preictal and interictal is also uncertain. Generally, researchers define the

preictal period as 5–30 min before seizure onset and the interictal period as between at least 2 or 4 h before seizure onset and 2 or 4 h after seizure end [36], [37], [38], [39]. The prediction algorithms are typically evaluated only using the defined preictal and interictal EEG samples, while all EEG samples between the preictal and interictal periods are discarded. Such a preprocessing method significantly reduces the available EEG samples. Furthermore, the continuity of the EEG recordings is disrupted, making the dataset obtained differ from the real data. Whether such datasets are suitable for evaluating the performance of seizure prediction methods remains to be discussed.

B. Epileptic Seizure Detection

The traditional EEG-based seizure detection algorithms focus on feature extraction, which has a substantial influence on the classification performance of conventional machine learning methods. The extracted features can be classified as time, frequency, time-frequency, or spatiotemporal domains. The statistical parameters of EEG signals are changed during seizures. Hence, time-domain features, e.g., approximate entropy [21] and Hurst exponent [40], are extracted to characterize EEG signals. To capture the frequency changes of the EEG signals during seizures, some works employ FFT [22] and discrete cosine transform (DCT) [23] to obtain frequency-domain features. Since the EEG signals are nonstationary in nature, time-frequency analysis is developed for the classification of epileptic seizures, where STFT [24] and discrete wavelet transform (DWT) [25] are widely used. The spatiotemporal domain analysis can detect the change of brain dynamics prior to seizure activity; spatiotemporal domain features have also been employed for automatic epileptic seizure detection, such as dynamical similarity [26], mean phase coherence [27], and phase-locking synchrony [28]. Those conventional machine learning methods have achieved promising results. However, manual feature extraction engineering relies on the experience of experts, which can be easily biased.

Recently, several studies have developed deep learning methods for seizure detection. In contrast to machine learning methods, deep learning models can automatically learn discriminative features from raw EEG signals without feature extraction. In [30], EEG segments are converted into 2-D images. 2D-CNNs are employed to extract features and classify seizure states. In [31], 3D-CNN is used to extract time-domain and spatial-domain features simultaneously. In [32], CNN is used as an autoencoder, while bidirectional long short-term memory (Bi-LSTM) is used as a classifier for automatic seizure detection. In [50], a 2-D convolutional LSTM model that combines CNN and LSTM in a single layer is used for seizure detection. These deep learning methods, however, still cannot fulfill the requirements for seizure alarming/closed-loop EEG systems. For example, in [10], a seizure detection processor is developed for closed-loop epilepsy management. To obtain a short detection latency, EEG recordings are split into nonoverlapping segments of 2 s, while segment-based outputs (i.e., 1800/h) are provided. Without postprocessing, the processor with 99.5%

segment-based specificity can still produce unacceptable numbers of false alarms ($\sim 9/h$). In [41] and [42], CNN is used as a seizure detector, while postprocessing is adopted to reduce the FAR. However, these postprocessing methods either degrade the sensitivity or introduce long detection latency (> 5 s).

Deep learning models have high classification performance, yet also have high-computational complexity and energy consumption, which is not conducive to being implemented on ultralow-power wearable devices. Lightweight seizure detection models are proposed in [43], [44], and [45] for low-power hardware implementations. These lightweight models, however, either have poor classification performance or require postprocessing to reduce the FAR. As mentioned in Section I, several multistage classification schemes are adopted to reduce the computing complexity [11], [12], [13], [14]. However, the cascaded coarse and fine classifiers introduce long detection latency, which has to be traded off with FAR.

To reduce the high computation complexity introduced by deep learning models, while maintaining high sensitivity, short detection latency, and low FAR simultaneously, is therefore the main target that we intend to achieve in this work.

III. METHODOLOGY

In this section, we propose a two-stage prediction + detection deep learning model, PDNet, for real-time epileptic seizure monitoring, which bears the merits of lightweight, high classification accuracy, high sensitivity, low FAR, and short detection latency. The proposed two-stage model PDNet consists of a prediction model and a detection model. The prediction model is used for the purpose of preprocessing instead of pure prediction and works together with the detection model to detect the occurrence of seizures.

The workflow and architecture of the proposed PDNet are presented in Section III-A. To improve the decision boundary between the preictal and interictal states, the proposed prediction model is trained by a semisupervised learning method, which is discussed in Section III-B. To precisely and rapidly classify the seizure states, the seizure detector is trained with soft labels, which are discussed in Section III-C. The training methodology for PDNet is discussed in Section III-D.

A. Two-Stage Prediction + Detection Model

The workflow and architecture of the proposed two-stage prediction + detection model PDNet is shown in Fig. 1. Only when the first-stage seizure predictor forecasts an impending seizure, the second-stage seizure detector is enabled to classify the seizure states. When the seizure terminates, the seizure detector is turned off and the seizure predictor is enabled to predict the next seizure.

There are three key points to make PDNet able to achieve high detection accuracy, low FAR, short detection latency, and lightweight computing simultaneously.

- 1) The seizure predictor with postprocessing is used for preprocessing of the seizure detection stage to reduce FAR, thereby avoiding the long detection latency introduced by postprocessing after seizure detection.

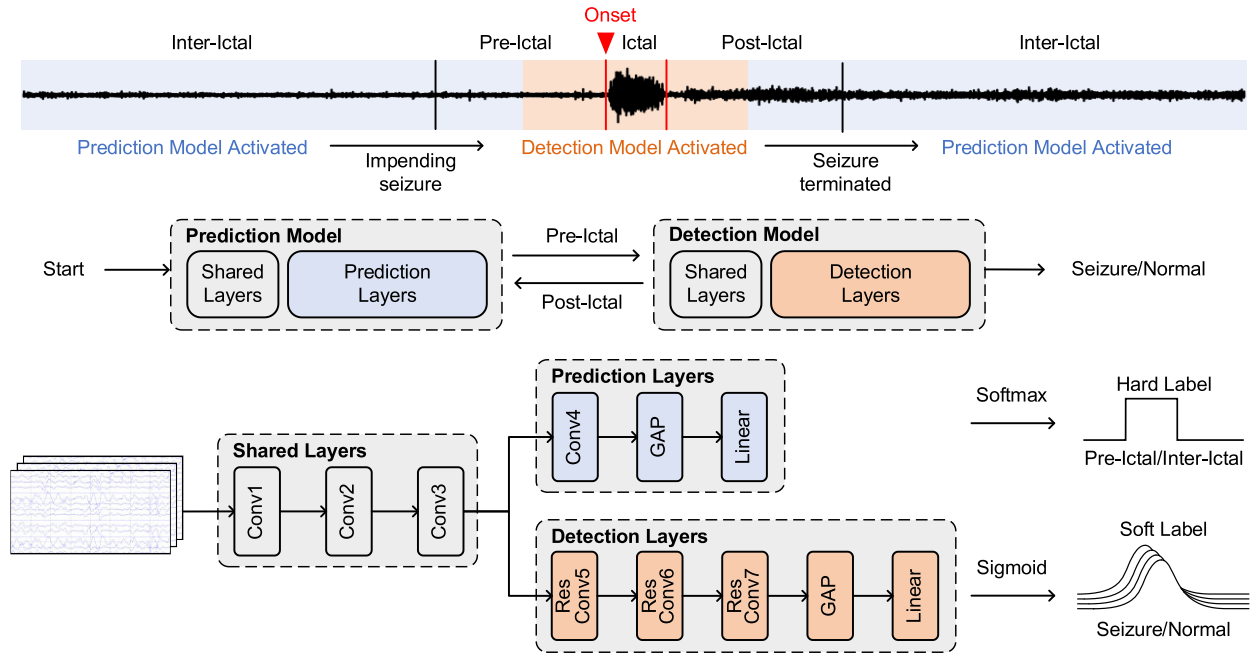


Fig. 1. Workflow and architecture of the proposed two-stage prediction + detection model PDNet. Conv: convolution layers. Res Conv: convolution layers with the residual connection.

- 2) Soft labels are employed to facilitate the proposed seizure monitor to precisely recognize the process of a seizure. The seizure detector with soft labels therefore achieves highly accurate classification.
- 3) Due to the low occurrence rate of seizures, the lightweight predictor dominates the runtime of the seizure monitoring framework, while the high-accuracy detector only works when necessary. Therefore, the PDNet framework achieves overall lightweight seizure monitoring, which is critical for the deployment of wearable edge devices.

In the proposed PDNet framework, the lightweight predictor is shallow and contains fewer parameters and computations. The prediction model and the detection model share the first few layers of the neural network, allowing them to share the learned features while reducing both parameters and computations. The high-accuracy detector is deep and with residual connection to prevent the gradient degradation issue [18]. Soft labels are employed to make the seizure detector more accurate in classifying the seizure states.

As shown in Fig. 1, PDNet is constructed by 1-D convolution, which can automatically learn features from the raw EEG data with less computation and parameters compared with 2-D and 3-D convolutions [45]. Leaky rectified linear unit (ReLU) is employed as the activation function for all the convolution layers. Following the convolution layers, a global average pooling (GAP) layer is used to shrink the activation maps for the linear layer. Finally, the probability output for the prediction model is generated by a softmax layer. Alternatively, the soft output for the detection model is generated by a sigmoid layer. The detailed architecture parameters of the proposed model are presented in Section IV-C.

B. Consistency-Based Semisupervised Seizure Prediction Model

As reviewed before, the boundary between the preictal and interictal states is uncertain. Setting a boundary and dividing the samples into preictal or interictal states arbitrarily leads to mislabeled samples. The disinformation introduced by the mislabeled samples greatly degrades the classification accuracy of supervised training. Previous works [36], [37], [38], [39] discard the EEG samples between the interictal and preictal states to avoid the disinformation. Such a segmentation method disrupts the continuity of the EEG data, making the dataset obtained differ from the real data. In this study, we retain the EEG samples between the preictal and interictal states as unlabeled samples and train the prediction model in a semisupervised setting.

Consistency Regularization methods are widely employed in semisupervised learning, with which discriminative features can be learned from unlabeled data [19]. Liang et al. [48] use the consistency regularization method to enhance the discriminability of the target patient by using extra unlabeled data. In the Consistency Regularization method, the inputs of models are perturbed by stochastic augmentations in each epoch. Consistency regularization allows the model to maintain consistent results for similar samples with minor variations. Therefore, the decision boundary is pushed to the low-density regions of the feature space as similar samples gather. Laine and Aila [20] present an efficient consistency regularization method, temporal ensembling, which ensembles the outputs of the network-in-training on different epochs as the prediction of the unknown labels under different perturbations. Temporal ensembling can provide resistance to disinformation and maintain a good performance even when a large portion

of samples are mislabeled. We adopt the temporal ensembling method as consistency regularization to further improve the decision boundary and enhance the tolerance to “incorrect labels” between the preictal and interictal states.

In this study, the interictal period is defined as between at least 2 h before seizure onset and 2 h after seizure end, which is the same as [36], [37], [38], and [39]. The preictal period is defined as between the interictal and ictal periods. We choose EEG samples from the preictal period, 5 min before seizure, as labeled data, since the data in this period probably contain information about the impending seizures. The other EEG samples in the preictal period are unlabeled data. Let D represent the whole training set. The EEG sample is denoted as x . The dataset for training the proposed prediction model D_p consists of two parts: the labeled data D_{pl} and the unlabeled data D_{pu} .

We adopt Gaussian augmentation (i.e., adding appropriate Gaussian noise to the inputs) and dropout as the stochastic augmentation in this study. Formally, the predicted vector of the proposed prediction model $\hat{y}_p$ is

$$\hat{y}_p = \mathcal{F}_p(G(x), \theta_p), \quad x \in D_p \quad (1)$$

where $\mathcal{F}_p(\cdot)$ denotes the function of the proposed prediction model, $G(\cdot)$ is the Gaussian augmentation function, and θ_p is the parameter set of the proposed prediction model. After each training epoch, the ensemble prediction Y_{ep} is constructed by accumulating the predicted vector $\hat{y}_p$, which is obtained as

$$Y_{ep} = mY_{ep} + (1 - m)\hat{y}_p \quad (2)$$

where m is the momentum term that controls the influence of the previous outputs. A factor $1/(1 - m^t)$ is used to scale the ensemble prediction to correct the start-up bias. The final ensemble prediction target y_{ep} is derived as

$$y_{ep} = \frac{Y_{ep}}{(1 - m^t)} \quad (3)$$

where t denotes the training epoch. The loss function of the semisupervised seizure prediction model consists of two parts. The first part is evaluated for the labeled data D_{pl} . We adopt focal loss [33] to address the imbalance between the positive (preictal data) samples and the negative samples (interictal data). The second part evaluates all the training data D_p . We adopt the mean square error (MSE) loss to penalize the bias between the predicted vector of the inputs with perturbations and the ensemble prediction.

Focal loss addresses the class imbalance by considering positive/negative samples and easy/hard samples, which is defined as

$$L_{fl} = \begin{cases} -\alpha(1 - \hat{y}_p)^\gamma \log \hat{y}_p, & y_p = 1 \\ -(1 - \alpha)\hat{y}_p^\gamma \log(1 - \hat{y}_p), & y_p = 0 \end{cases} \quad (4)$$

where y_p denotes the ground-truth label, $\hat{y}_p$ denotes the predicted output, α is the weighting factor for the importance of positive/negative samples, and γ is the modulating factor for easy/hard samples. The modulating factor downweighed the loss of well-classified samples to make the model focus

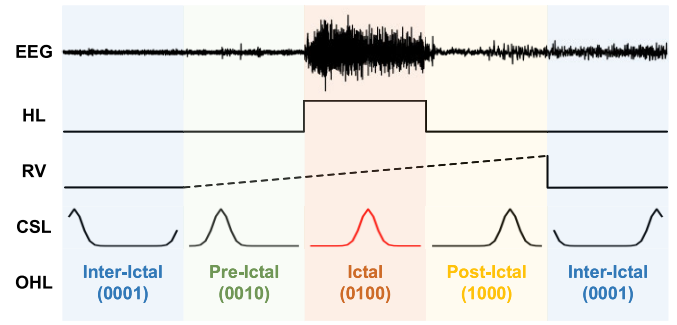


Fig. 2. Illustration of seizure states represented by HL, RVs, CSL, and OHL.

on the hard samples. The mean square error loss is given as

$$L_{mse} = \frac{1}{N} \sum \|y_{ep} - \hat{y}_p\|^2. \quad (5)$$

Since the ensemble prediction y_{ep} takes several epochs to converge, the confidence of the prediction of unlabeled data is deficient in the beginning. To avoid ambiguous labels that may interfere with the training, unsupervised loss L_{mse} is weighted using a ramp-up weighting function $\omega(t)$. The ramp-up weighting function is set to zero on the first training epoch and gradually increases. The loss function of the proposed prediction model is given as $L_p = L_{fl} + \omega(t)L_{mse}$. The ramp-up weighting function $\omega(t)$ is defined as

$$\omega(t) = \begin{cases} \omega_{max} \times \exp\left[-5\left(1 - \frac{t}{t_{th}}\right)^2\right], & t \leq t_{th} \\ \omega_{max}, & t > t_{th} \end{cases} \quad (6)$$

where ω_{max} is the scale factor of the ramp-up weighting function. Note that the division of interictal and preictal data are only employed during training. In the inference process, all the test data are evaluated.

C. Seizure Detection Model With Soft Labels

Though the EEG recordings can be categorized into four states (i.e., interictal, preictal, ictal, and postictal), the existing seizure detection studies usually focus on binary classification. HLs are used to represent seizure and nonseizure states, which leads to the loss of information in the EEG recordings. To precisely distinguish the different stages of a seizure, the proposed seizure detection model performs a multicategory task.

In general, researchers use one-hot labels (OHLs) for multicategory tasks or convert classification tasks to regression problems. In Fig. 2, the four seizure states and the labels represented by HL, regression value (RV), soft label-CSL, and OHL are shown. The OHLs cannot represent the connection between the adjacent categories. The RV would be discontinuous at the boundary (when the postictal state changes back to the interictal state). The labels representing seizure states should maintain the continuity at the boundaries between adjacent states.

The CSLs exhibit great success in angle prediction of rotated object detection in aerial images for two properties, periodicity and smoothness [34]. The periodicity property can

handle the discontinuity at the boundary, while the smoothness property helps reduce penalties when samples are misclassified into the categories adjacent to the ground truth. We adopt CSLs to make the detection model learn from the connections between different seizure states and avoid discontinuity at the boundaries.

The CSLs used to train the detection model is based on Gaussian distributions. If a task contains n_c categories, the smooth label is a n_c -dimension vector. The smooth label for the c th category is given as

$$y_g(c) = \begin{cases} \exp\left(-\frac{n_t^2}{2\sigma^2}\right), & n_t < \left[\frac{n_c}{2}\right] \\ \exp\left(-\frac{(n_t - n_c)^2}{2\sigma^2}\right), & n_t \geq \left[\frac{n_c}{2}\right] \end{cases} \quad (7)$$

where $n_t = n - c$, $n \in [0, n_c]$, $n \in \mathbb{Z}$. σ is the standard deviation of Gaussian distribution. As shown in Fig. 2, CSLs can be regarded as cyclic shifting the Gaussian distributions to the central c . The position of the center from left to right indicates the seizure state cycle “interictal–preictal–ictal–postictal–interictal.” The Gaussian distributions of adjacent categories have overlapping parts, as well as the first and last categories. The CSLs show smoothness and periodicity. The postictal period usually lasts 5–30 min [47]. Since we are more interested in the state transitions from preictal–ictal–postictal, we choose short preictal and postictal periods (i.e., 5 min) for the soft labels. In the study, the soft labels contain a total of 11 categories, where $[0, 10]$, $[1, 2, 3]$, $[4, 5, 6]$, and $[7, 8, 9]$ represent interictal state, preict state, ictal state, and postictal state, respectively. The choice of 11 categories is a tradeoff between the classification accuracy for seizure states and task difficulty. Each state categorization is evenly distributed by timestamp. For example, we use $[7, 8, 9]$ to represent a 5-min postictal period, i.e., 7, 8, and 9 represent the samples in the first, middle, and the last 100 s, respectively.

The proposed detection model is trained with CSLs y_d . The predicted seizure vector of the proposed detection model $\hat{y}_d$ is derived as

$$\hat{y}_d = \mathcal{F}_d(x, \theta_d), \quad x \in D \quad (8)$$

where $\mathcal{F}_d(\cdot)$ is the function of the proposed detection model. θ_d denotes the parameters of the proposed detection model. We adopt Gaussian focal loss [34] as the objective function of our proposed detection model. Gaussian focal loss is the summation of n_c -dimension vector. The loss function of the proposed detection model L_d , i.e., Gaussian focal loss is given as

$$L_d = L_{\text{gfl}} = -\frac{1}{n_c} \sum_{n \in n_c} \begin{cases} (1 - n)^\lambda \log n, & y_d = 1 \\ -(1 - y_d)^\beta n^\lambda \log(1 - n), & y_d \neq 1 \end{cases} \quad (9)$$

where y_d denotes the ground-truth label, and β is the Gaussian reweight factor, which provides error tolerance for the prediction close to the central. λ is the modulating factor, which is used to focus the training on central c and downweigh the values of other positions.

TABLE I
DATA INFORMATION OF CHB-MIT DATABASE

Patient	Age	Gender	No. of Seizure	EEG duration	Patient	Age	Gender	No. of Seizure	EEG duration
1	11	F	7	40.5	11	12	F	3	34.8
2	11	M	3	35.3	14	9	F	6	23.2
3	14	F	7	38.0	16	7	F	5	16.1
4	22	M	4	156.1	17	12	F	3	20.0
5	7	F	5	39.0	18	18	F	5	33.7
6	1.5	F	10	66.7	19	19	F	3	28.9
7	14.5	F	3	67.0	20	6	F	6	26.3
8	3.5	M	5	20.0	21	13	F	4	32.8
9	10	F	4	67.9	22	9	F	3	31.0
10	3	M	7	50.0	23	6	F	5	23.9

D. Training Methodology

Since the proposed prediction model and detection model share the first few layers of the network, the two models are trained simultaneously. The training process of the proposed two-stage model is shown in Fig. 3, while the pseudocode is shown in Algorithm 1. In the proposed approach, the loss function of the two-stage model consists of two main components: L_p and L_d . L_p denotes the loss function of the prediction model, which consists of the focal loss L_{fl} and the mse loss L_{mse} . L_d denotes the loss function of the detection model, i.e., Gaussian focal loss L_{gfl} . We combine the loss of the prediction model and detection model to obtain the final loss function L of the proposed training approach, which is

$$L = L_p + L_d = L_{\text{fl}} + \omega(t)L_{\text{mse}} + L_{\text{gfl}}. \quad (10)$$

The detailed setting of hyperparameters is explained in Section IV-C.

IV. EXPERIMENT SETUP AND FLOW

A. Dataset

The EEG data used in this article is acquired from the CHB-MIT scalp dataset [9], which consists of long-term EEG recordings from 23 children with intractable seizures. In the CHB-MIT database, EEG data are recorded according to the international 10–20 electrode system at a sampling rate of 256 Hz.

Following other works [36], [37], [38], [39], we define the interictal period as between at least 2 h before seizure onset and 2 h after seizure end. The preictal period is defined as between the interictal and ictal periods. In the case of two seizures occurring at a short interval, if the interval is less than 30 min, we combine the second seizure with the leading one as one seizure. To avoid overfitting, patients with at least a 3-h interictal period are selected for the evaluation of performance. In summary, the remaining database contains ~851 h of EEG data, including 98 seizure events in 20 patients. The details about the used CHB-MIT database are given in Table I.

B. Preprocessing

The preprocessing step includes EEG channel selection, digital filtering, and window segmentation. The EEG data in the CHB-MIT dataset use different electrodes. To ensure the

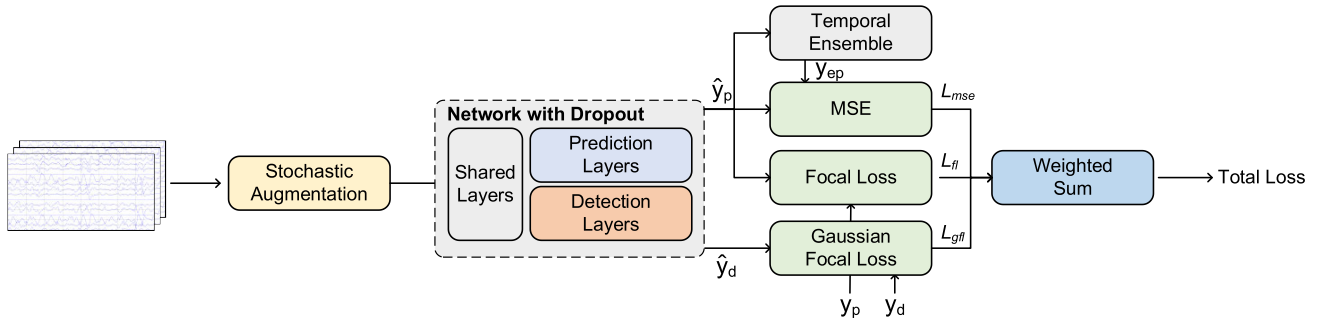


Fig. 3. Overall training process of the proposed two-stage model. $\hat{y}_p$ denotes the output of the prediction model. $\hat{y}_d$ denotes the output of the detection model. y_{ep} denotes the ensemble prediction vectors. y_p denotes labels for labeled samples for the prediction model. y_d denotes labels for the detection model.

Algorithm 1 The Proposed Training Process

Require: x_i = training data
Require: D_{pl} = training set of known labels for prediction model; D_{pu} = training set of unknown labels for prediction model; D_{dl} = training set of known labels for detection model
Require: y_p^i = hard labels for labeled inputs $i \in D_{pl}$; y_d^i = soft labels for labeled inputs $i \in D_{dl}$
Require: m = ensembling momentum; α = weighting factor; γ = modulating factor
Require: β = Gaussian re-weight factor; λ = modulating factor of soft label
Require: $\omega(t)$ = unsupervised weight ramp-up function
Require: $F_\theta(x)$ = the proposed two-stage model with trainable parameters θ
Require: $F_{mse}(x)$ = mean square error loss function
Require: $F_{fl}(x)$ = focal loss function
Require: $F_{gfl}(x)$ = Gaussian focal loss function
Require: $G(x)$ = stochastic input augmentation function

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 $Y_{ep} \leftarrow \mathbf{0}$                                 ▷ initialize ensemble predictions
 $y_{ep} \leftarrow \mathbf{0}$                             ▷ initialize target vectors
for  $t$  in  $[1, \text{num\_epochs}]$  do
    for each mini-batch  $B$  do
         $\hat{y}_p^i, \hat{y}_d^i \leftarrow F_\theta(G(x_i)), i \in B$                 ▷ evaluate outputs for augmented inputs
         $L_{mse} \leftarrow F_{mse}(y_{ep}^i, \hat{y}_p^i), i \in B \cap D_{pu}$         ▷ unsupervised loss component
         $L_{fl} \leftarrow F_{fl}(y_p^i, \hat{y}_p^i, \alpha, \gamma), i \in B \cap D_{pl}$     ▷ supervised loss component of hard labels
         $L_{gfl} \leftarrow F_{gfl}(y_d^i, \hat{y}_d^i, \beta, \lambda), i \in B \cap D_{dl}$     ▷ supervised loss component of soft labels
         $L \leftarrow L_{fl} + \omega(t)L_{mse} + L_{gfl}$                 ▷ total loss of the two-stage model
        Update  $\theta$ , using, e.g., ADAM
    end for
     $Y_{ep} \leftarrow mY_{ep} + (1 - m)\hat{y}_p$                 ▷ accumulate ensemble predictions
     $y_{ep} = Y_{ep} / (1 - m^t)$                 ▷ construct target vectors by bias correction
end for
return  $\theta$ 
    
```

consistency of the channel configuration, we select 16 channels for all the patients, including FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, and P8-O2. To attenuate the interference caused by the high-frequency noise and power line noise, the raw EEG recordings are denoised by a 10th-order Butterworth bandpass filter (0.5–45 Hz) and then split into nonoverlapping windows of 2 s (16×512 points). Since the proportion of the length of seizure is typically less than 2% of the entire EEG recordings for epilepsy patients, the numbers of positive and negative samples in the obtained dataset are highly imbalanced. We address the imbalance issue by using focal loss and resampling techniques, which are discussed in Section IV-C.

C. Training Details

The experiments are based on Python 3.10.13 and PyTorch 2.1.0 framework. The detailed structure parameters of the proposed two-stage PDNet model are given in Table II. We adopt the Adam optimizer with a batch size of 256 and an initial learning rate of 0.001. A dropout rate of 0.5 is added to the prediction layers and detection layers to provide stochastic perturbations and prevent overfitting. The ensembling momentum m and modulating factor γ are 0.6 and 2.0, respectively. The Gaussian reweight factor β and modulating factor of soft label λ are 1.5 and 3.0, respectively. The parameters are set based on the recommended settings by focal loss [33], Gaussian focal loss [34], and empirical hyperparameter tuning.

TABLE II
STRUCTURE PARAMETERS OF THE PROPOSED PDNET MODEL

Layer Name	Shared Layers			
	Output size	Kernel size		
Input	(16, 512)	—		
Conv1	(16, 8, 512)	$8 \times 1 \times 5$		
MaxPool	(16, 8, 256)	$1 \times 1 \times 2$		
Conv2	(16, 16, 256)	$\begin{bmatrix} 2 \times 8 \times 1 \\ 16 \times 2 \times 5 \end{bmatrix}$		
MaxPool	(16, 16, 128)	$1 \times 1 \times 2$		
Conv3	(16, 16, 128)	$\begin{bmatrix} 4 \times 16 \times 1 \\ 16 \times 4 \times 5 \end{bmatrix}$		
Flatten	(256, 128)	—		
Layer Name	Prediction Layers		Detection Layers	
	Output size	Kernel size	Output size	Kernel size
Conv4	(64, 128)	$\begin{bmatrix} 8 \times 256 \times 1 \\ 64 \times 8 \times 5 \end{bmatrix}$	—	—
Res Conv5 [#]	—	—	(32, 128)	$\begin{bmatrix} 16 \times 256 \times 1 \\ 16 \times 16 \times 5 \\ 32 \times 16 \times 1 \end{bmatrix}$
Res Conv6 [#]	—	—	(48, 128)	$\begin{bmatrix} 32 \times 32 \times 1 \\ 32 \times 32 \times 5 \\ 48 \times 32 \times 1 \end{bmatrix}$
Res Conv7 [#]	—	—	(64, 128)	$\begin{bmatrix} 48 \times 48 \times 1 \\ 48 \times 48 \times 5 \\ 64 \times 48 \times 1 \end{bmatrix}$
GAP	64	—	64	—
Linear	2	2×64	11	11×64

[#]: Res Conv denotes convolution with residual connection.

For the training of the proposed prediction model, we adopt focal loss to address the imbalance between positive samples (preictal data) and negative samples (interictal data). The weighting factor α is set to the ratio of positive samples in the training set, which balances the total loss of positive/negative samples. Thus, the data distribution of positive samples and negative samples in the training of the prediction model is consistent with that in practice.

For the training of the proposed detection model, the numbers of positive samples (ictal data) and negative samples (nonictal data) are also highly skewed. Since the detection model is designed to be active only when the prediction model detects an impending seizure, the data distribution of positive and negative samples that the detection model runs against is different from the data distribution of the entire dataset. Note that the modulating factor λ of Gaussian focal loss L_{gfl} is used to reweight the loss of positive position (central) and negative position within the circular labels, but not to reweight the loss of positive/negative samples. The focal loss L_{fl} that is used for the training of the prediction model is not used to address the imbalance issue for the detection model. In this study, we obtain a balanced dataset for training the detection model by resampling. In each epoch, we randomly discard some of the negative samples from the training set to make the number of negative samples and positive samples equal. Note that the resampling operation is not used in the evaluation, which is only used to obtain a balanced training subset to facilitate the training of the detection model.

D. Postprocessing of the Prediction Stage

Generally, postprocessing methods are used to reduce the FAR. Postprocessing, however, easily introduces long latency, which is unacceptable for seizure detection, whereas is fine for

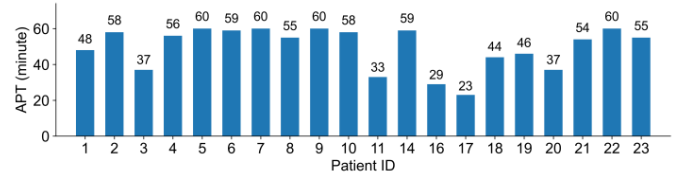


Fig. 4. APT of all patients.

seizure prediction. In this study, we only adopt postprocessing method to the prediction model, but not to the detection model. We adopt k -of- n strategy as [37], where an alarm is triggered only if at least k outputs during the last n outputs are classified to be preictal. Specifically, if at least 18 s of 30-s consecutive signals are classified as positive, this is considered a positive event. The choice of k and n is a trade-off between sensitivity and FAR.

E. Performance Evaluation

To obtain an unbiased evaluation of the proposed model, we choose a patient-specific leave-one-out cross-validation (LOOCV) strategy to verify the proposed model in real scenarios. Specifically, suppose there are N seizures in the dataset for a certain patient. In each leave-one-out loop, one seizure is retained as a test set while the remaining $(N - 1)$ seizures are used for training. This procedure repeats N times till all the N seizures of the patient have been selected as a test set for evaluation. The performance metrics are reported based on the average across N loops.

Because the exact time of seizure is unpredictable, the seizure occurrence period (SOP) is proposed to evaluate the seizure prediction method [27]. SOP is the period when the seizure is expected to occur, the length of which is equal to the length of the preictal period. In this study, we use an adaptive SOP as [39] instead of using a fixed SOP for all patients as in other studies [35], [36], [37], [38]. In each leave-one-out loop, the SOP for the testing seizure is set as the average prediction time (APT) of the $(N-1)$ seizures used for training, with a maximum of 1 h. The prediction time is defined as the time from alarm to seizure onset. The APT for all patients is shown in Fig. 4. The APT is also the expected time from alarm to seizure onset for the patient. The patient-specific APT can help the patient anticipate the time of an upcoming seizure, thereby reducing psychological stress. Caregivers or clinicians can also provide customized healthcare and treatment for patients according to the patient-specific seizure prediction time.

For the proposed prediction model, we evaluate event-based sensitivity (Sen), FAR, time in warning ρ_w , and p-value. Specifically, ρ_w denotes the ratio of time spent in warning (from alarm to seizure onset) to total time. p-value [29] is used to measure whether the proposed seizure predictor is statistically better than a random predictor.

For the proposed detection model, we evaluate event-based sensitivity, FAR, and detection latency T_d . Specifically, T_d is the time interval between the seizure onset and the raised alarm.

TABLE III
 RESULTS OF THE PROPOSED TWO-STAGE MODEL

Patient	Prediction Model				Detection Model			Two-Stage Model		
	Sen (%)	ρ_w (%)	FAR/h	p-value	Sen (%)	FAR/h	T_d /s	Sen (%)	FAR/h	T_d /s
1	100.0	15.9	0.022	<0.001	100.0	5.28	1.14	100.0	0.68	1.14
2	100.0	6.4	0.000	<0.001	100.0	1.22	4.33	100.0	0.13	4.33
3	100.0	9.5	0.000	<0.001	100.0	2.40	0.71	100.0	0.72	0.71
4	100.0	2.5	0.000	<0.001	100.0	5.20	5.50	100.0	0.13	5.50
5	80.0	15.0	0.028	0.002	100.0	0.22	3.40	100.0	0.03	9.60
6	100.0	18.5	0.027	<0.001	100.0	0.73	1.10	100.0	0.11	1.10
7	100.0	7.3	0.000	<0.001	100.0	0.54	1.33	100.0	0.08	1.33
8	100.0	23.6	0.000	0.001	100.0	5.61	2.40	100.0	0.93	2.40
9	75.0	9.1	0.016	0.001	100.0	0.65	5.50	100.0	0.05	11.25
10	85.7	12.7	0.000	<0.001	100.0	5.83	0.71	100.0	1.16	3.57
11	100.0	3.4	0.000	<0.001	100.0	1.99	0.33	100.0	0.54	0.33
14	100.0	26.7	0.000	<0.001	100.0	2.20	2.00	100.0	1.21	2.00
16	80.0	15.0	0.000	0.002	80.0	5.84	1.00	80.0	0.89	1.00
17	100.0	5.2	0.000	<0.001	100.0	4.52	4.67	100.0	0.33	4.67
18	80.0	6.3	0.026	<0.001	100.0	2.64	2.40	100.0	0.13	11.20
19	100.0	4.1	0.000	<0.001	100.0	4.21	2.00	100.0	0.18	2.00
20	100.0	14.5	0.000	<0.001	100.0	2.97	3.67	100.0	0.08	3.67
21	100.0	11.6	0.000	<0.001	100.0	3.67	1.50	100.0	0.52	1.50
22	100.0	12.8	0.000	0.002	100.0	0.34	2.33	100.0	0.11	2.33
23	100.0	18.9	0.000	<0.001	100.0	4.05	0.80	100.0	0.62	0.80
Average	95.0	11.9	0.006	—	99.0	3.01	2.34	99.0	0.43	3.52

V. RESULTS

A. Efficacy of Two-Stage Classification Scheme

In this work, we evaluate the proposed two-stage model on the CHB-MIT EEG dataset using LOOCV. The results of the proposed two-stage model are presented in Table III. When running alone, the proposed seizure prediction model achieves an average sensitivity of 95.0% and an average FAR of 0.006/h for prediction purposes. The proposed seizure detection model achieves an average sensitivity of 99.0%, an average FAR of 3.01/h, and an average detection latency of 2.34 s for detection purposes. When running the proposed prediction and detection models together for seizure detection purposes, the proposed two-stage model achieves an average sensitivity of 99.0%, an average FAR of 0.43/h, and an average detection latency of 3.52 s. Note that there are four seizures (one each for Patients 5, 9, 10, and 18) that are not predicted in the preictal state by the prediction model, yet “predicted” in the ictal state. Then, the detection model is activated and captures these four seizures. In this case, the proposed two-stage model maintains the high sensitivity to seizure events in the trade of a long detection latency. The results demonstrate the error tolerance of the proposed two-stage monitoring scheme, even when there are missing predictions of seizures by the prediction model in the preictal stage.

In terms of complexity, the lightweight prediction model contains 5.34k parameters and 2.43M multiply-accumulates (MACs), while the highly accurate detection model contains 44.57k parameters and 7.37M MACs. The runtime of the seizure prediction model and seizure detection model is illustrated in Fig. 5. The prediction model and the detection model are activated alternatively. The detection model is only

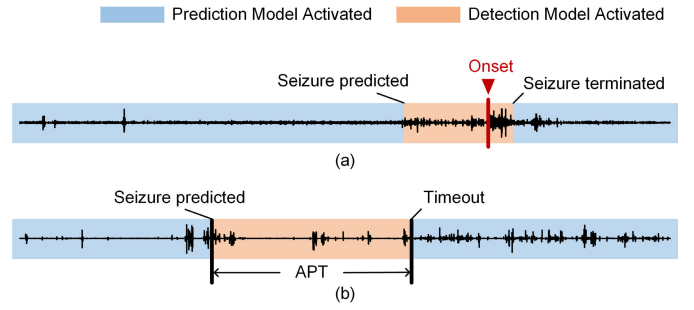


Fig. 5. Illustration of the runtime of seizure prediction model and seizure detection model. (a) Seizure is accurately predicted. (b) False alarm.

activated when the prediction model triggers an alarm and closes when the seizure terminates. If the predicted seizure is a false alarm, the detection model is closed after the APT. The average runtime of the highly accurate detection model is approximately equal to the average time in warning ρ_w ($\sim 12\%$). The average runtime of the lightweight prediction model is approximately equal to $1 - \rho_w$ ($\sim 88\%$). The two-stage classification scheme contains an average computation of 3.02M MACs based on the average runtime of the prediction and detection models. Compared with using the proposed detection model alone, the two-stage scheme reduces the computation complexity by 59.0% and FAR by 85.6%, with no sensitivity drop.

B. Comparisons for Seizure Prediction Performance

We also compare the proposed prediction model alone with the state-of-the-art prediction methods to evaluate the performance of seizure prediction. The results are summarized in Table IV. Note that only the methods evaluated with LOOCV and the same database are selected for comparison.

Compared with the methods combining feature extraction methods and deep learning models [35], [36], the proposed seizure prediction model has significant advantages in sensitivity and FAR. Specifically, compared to [35] and [36], our proposed prediction model enhances the sensitivity by 1.5% and 9.3%, and reduces the FAR by 95.9% and 93.8%, respectively.

Compared with the methods only using deep learning models [37], [38], [39], the proposed seizure prediction model significantly reduces the FAR and the number of calculations. Specifically, compared to [37], [38], and [39], the proposed prediction model reduces the FAR by 81.8%, 92.2%, and 94.5%, and the amount of calculation by $12.4\times$, $3.1\times$, and $51.2\times$, respectively. In terms of sensitivity, compared to [37] and [38], the proposed prediction model enhances the sensitivity by 1.9% and 0.1%, respectively. Compared to the proposed model, [39] enhances the sensitivity by 0.5%, yet at the cost of high computation complexity.

C. Comparisons for Seizure Detection Performance

The comparisons of our proposed two-stage model for detection purposes with other methods are summarized in Table V. Again, note that only the methods evaluated with

TABLE IV
COMPARISONS OF THE PROPOSED PREDICTION MODEL TO THE STATE OF THE ART

Index	Methods	Validation strategy	No. of patients / seizures	Inter-ictal / pre-ictal intervals (min)	Evaluated hours	Sen (%)	FAR (/h)	Parameters	MACs
[35]	STFT+CNN	LOOCV	23 / 156	240 / 30	296.8	93.5	0.145	—	—
[36]	Spectral power, statistical moment, Hjorth+CNN	LOOCV	16 / 77	60 / 60 120 / 60 240 / 60	466.1 419.4 353.5	86.8 87.0 85.7	0.292 0.186 0.096	—	—
[37]	raw+CNN	LOOCV	18 / 87	240 / 30	337.2	93.1	0.033	163k	30M
[38]	raw+CNN	LOOCV	19 / 105	240 / 30 240 / 60	426.5 461.4	93.0 94.9	0.094 0.077	116k	7.5M*
[39]	raw+GCN	LOOCV	19 / 98	120 / Adaptive	453.0	95.5	0.109	341k	124.4M
This work	raw+CNN	LOOCV	20 / 98	Adaptive	851.2	95.0	0.006	5.34k	2.43M

LOOCV: leave-one-out cross validation; CSP: common spatial pattern; GCN: graph convolutional network.

*: 7.5M multiplications and 54.3M additions.

TABLE V
COMPARISONS OF THE PROPOSED PDNET TO THE STATE-OF-THE-ART

Index	Methods [#]	Prediction	Post-processing	Detection latency (s)	Sen (%)	FAR (/h)	Parameters	MACs
[14]	raw+CNN+ three-stage classification	No	No	—	97.8	0.50	—	222.6k
[40]	temporal feature+ EasyEnsemble	No	two successive epochs	>5	97.5	0.91	—	—
[41]	raw+CNN	No	successive epochs + MTI	8.1	99.3	0.2	105.54k	8.23M
[42]	raw+CNN+ ScoreNet	No	ScoreNet	<10	92.0	0.09	17.3k [†]	6.35M [‡]
[43]	raw+CNN	No	No	5.83	91.4	1.99	17.3k	6.35M
[44]	temporal feature+ CNN	No	No	1.39	98.9	—	289.0k [†]	8.45M [‡]
This work	raw+CNN	No	No	2.34	99.0	3.01	44.57k	7.37M
		Yes	No	3.52	99.0	0.43	49.30k	3.02M [‡]

[#]: All the methods are evaluated with LOOCV except [14]. [†]: Only the parameters and the computations of the CNN model are counted.

[‡]: Estimated based on average runtime of prediction model and detection model.

the same dataset and LOOCV (except [14]) are included in the comparisons.

In [14], a seizure detection processor using a three-stage detection model is proposed, which is evaluated with the CHB-MIT dataset. However, it is not clear whether LOOCV is used for validation in [14]. This three-stage model achieves 97.78% sensitivity and 0.5/h FAR. Compared to [14], the proposed two-stage model enhances the sensitivity by 1.22% and reduces the FAR by 0.07/h. Furthermore, the detection latency of the three-stage model in [14] is not reported, which could be the bottleneck of this model.

Compared with other seizure detection models with post-processing [40], [41], [42], the proposed two-stage model achieves shorter detection latency and competitive classification performance in terms of sensitivity and FAR. Specifically, compared to [40], the proposed two-stage model enhances the sensitivity by 1.5% and reduces the FAR by 0.48/h, while reducing the detection latency by 29.6%. Compared to [41], the proposed model reduces the sensitivity by 0.3% and increases the FAR by 0.23/h, while reducing the detection

latency by 56.5%. Compared to [42], the proposed model enhances the sensitivity by 7.0% and increases the FAR by 0.34/h, while reducing the detection latency by 64.8%. Guo et al. [40] report the performance of their methods with and without postprocessing. The postprocessing method in [40] reduces the FAR from 1.38/h to 0.91/h, yet reduces the sensitivity by 2.5%. For the application requiring extremely low FAR, we can adopt postprocessing to further reduce FAR. For example, if we adopt the postprocessing method as [40] to trigger the alarm only when two successive epochs are classified positive, the sensitivity remains unchanged, the FAR would be reduced to 0.14/h, and the detection latency would be increased to 5.32 s.

Compared with other seizure detection models without post-processing [43], [44], the proposed two-stage method achieves higher sensitivity and lower FAR with competitive detection latency. Specifically, compared to [43], the proposed two-stage model enhances the sensitivity by 7.6%, reduces the detection latency by 2.22 s, and reduces the FAR by 78.4%. Compared to [44], the proposed two-stage model enhances the sensitivity

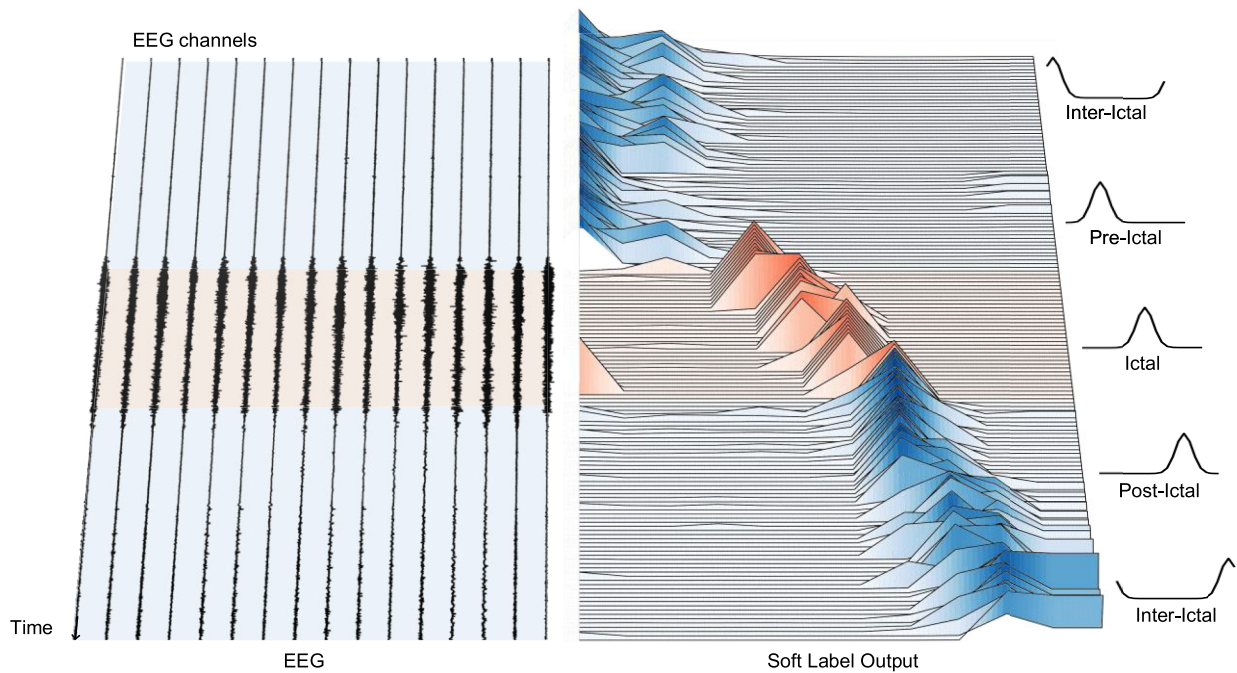


Fig. 6. Example of the soft label output and the EEG recordings of Patient 9. Each line represents the soft label output of one EEG sample.

by 0.1%, while increasing the detection latency by 2.13 s. The FAR of [44] is not reported. Their method achieves an average specificity of 96.05%, which implies a large FAR.

We also compare the number of computations with the proposed two-stage model and other methods with the deep learning model [41], [42], [43], [44] for the overall monitoring process. Note that [42], [43] use the same CNN model for feature extraction and classification. Compared to [41] and [42] (same for [43]), and [44], the proposed two-stage model reduces the amount of calculation by 63.3%, 52.4%, and 64.3%, respectively.

D. Efficacy of Soft Label

In this study, we introduce the soft label (CSLs) instead of the conventional HL to facilitate our proposed PDNet to classify the seizure states more precisely. To verify the effectiveness of the proposed soft label, we compare the performance of PDNet that is trained with the conventional HL and the soft label, respectively. The PDNet model trained with the conventional HL achieves 96.3% sensitivity, 0.99/h FAR, and 3.91-s detection latency. Compared with the model trained with HL, the model trained with soft label enhances the sensitivity by 2.7%, reduces the FAR by 0.56/h, and reduces the detection latency by 0.39 s.

The soft label also enables the model to classify the seizure states more precisely. An example of the soft label outputs and the EEG recordings of Patient 9 in the CHB-MIT dataset are shown in Fig. 6. Each line represents the soft label output of one EEG sample. As shown in the figure, the wave peak of the soft label output moves from the left to the right as the seizure proceeds from onset to termination in the EEG recording. Since no predictive characteristic or preictal biomarker is universal among people with epilepsy [6],

accurately categorizing the preictal state in each EEG sample is not possible. Note that the soft label outputs of the EEG samples in the preictal state are often categorized as interictal state as shown in Fig. 6. Benefited from the error tolerance introduced by the Gaussian focal loss, the misclassification of preictal and interictal states does not impair the performance of the detection model. Alternatively, the soft labels help to categorize the ictal and postictal states more precisely, thereby helping the patients better understand the process of a seizure, relieve their stress, and facilitate caregivers or clinicians to come up with targeted care and treatment.

E. Generalizability Evaluation

To validate the generalizability of our proposed model, we evaluate the PDNet model on another open-access database, Siena Scalp EEG Database [51]. The Siena Scalp EEG Database consists of long-term EEG recordings from 14 patients with epileptic seizures. In the Siena Scalp database, EEG data are also recorded according to the international 10–20 electrode system at a sampling rate of 512 Hz. We use the same criterion to select the used EEG recordings from the CHB-MIT database as described in Section IV-A. The remaining EEG recordings of the Siena database contain ~88 h of EEG data, including 32 seizure events from eight patients.

The preprocessing is the same as for the CHB-MIT database. We select 16 channels of Siena database for all the patients, including FP1-F7, F7-FC5, FC5-CP5, CP5-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-FC6, FC6-CP6, and CP6-O2. The raw EEG recordings are denoised by a 10th-order Butterworth bandpass filter (0.5–45 Hz) and then split into nonoverlapping windows of 2 s (16×1024 points).

When running alone, the proposed seizure prediction model achieves an average sensitivity of 93.1% and an average FAR of 0.008/h for prediction. The proposed seizure detection model achieves an average sensitivity of 96.9%, an average FAR of 10.4/h, and an average detection latency of 3.15 s for detection. Alternatively, when running the proposed prediction and detection models together for seizure detection purposes, the proposed two-stage PDNet model achieves an average sensitivity of 96.9%, an average FAR of 1.32/h, and an average detection latency of 3.64 s. The results of the Siena database demonstrate the generalizability of the PDNet model.

F. Deployment Procedure and Application Scenario

The proposed PDNet framework is designed for long-term epileptic seizure monitoring on wearable devices. Benefitting from the minimal number of parameters and computations, the proposed PDNet model can be easily deployed on ultralow-power processors of wearable devices to achieve long-term monitoring. To deploy the proposed PDNet model, long-term EEG recordings need to be collected first to obtain sufficient cases of seizures to train the model. Subsequently, the processors can be configured to deploy the PDNet model by loading the trained parameters.

With high sensitivity to seizure events, wearable devices with the proposed PDNet model can be used for long-term EEG visual inspection and treatment monitoring to significantly reduce the workload for clinicians. In addition, the proposed PDNet model can detect seizures at the initial stage, enabling the wearable devices to send early warnings to caregivers or to inform the implantable neural stimulation module to deliver responsive electrical stimulation to realize ambulatory epilepsy treatment. Furthermore, fine-grained information such as the occurrence process of seizures demonstrated by soft labels can help the caregivers or clinicians to come up with targeted healthcare and clinical treatments.

G. Limitations and Future Work

Although the proposed two-stage prediction + detection framework provides several advantages for epileptic seizure monitoring, two limitations remain in the current work. First, we select identical EEG channels for different patients in the dataset to construct the PDNet model. However, previous works prove that removing the EEG data from certain electrodes for different patients [14] may only lead to marginal performance loss for seizure detection. To further reduce the amount of computation, we could focus on patient-adaptive EEG channel selection in the future. Second, we propose a patient-specific model PDNet in this work, with the expectation that fine-grained information such as APT and the process of seizures demonstrated by soft labels can contribute to the development of personalized healthcare and clinical treatment strategies. However, long-term EEG recordings and seizure annotations must be collected first to train a patient-specific model. This limits the application of the proposed method. Therefore, we consider developing a patient-independent model in future work as a choice to further promote our two-stage framework.

VI. CONCLUSION

In this work, we propose a two-stage prediction + detection framework, PDNet, for real-time epileptic seizure monitoring, which achieves high detection accuracy, low FAR, short detection latency, and lightweight computing simultaneously. The proposed two-stage PDNet model consists of a lightweight seizure predictor and a highly accurate seizure detector. Only when the first-stage seizure predictor forecasts an impending seizure, the second-stage seizure detector is activated to precisely and rapidly classify the seizure states. The proposed seizure predictor with semisupervised learning effectively enhances the decision boundary, acting as a key enabler for the whole model to achieve low FAR, short detection latency, and minimal computations. Soft labels (CSLs) are adopted to allow the seizure detector to distinguish the different stages of a seizure, increasing the classification accuracy.

The proposed PDNet is evaluated using the CHB-MIT scalp EEG database. When running alone, the proposed prediction model achieves 95.0% sensitivity, 0.006/h FAR, with 2.43 M MACs. When running the proposed prediction and detection models together for seizure detection purposes, the PDNet achieves 99.0% sensitivity, 0.43/h FAR, and 3.52-s detection latency with 3.02M MACs, which are competitive compared to the state of the art in terms of sensitivity, detection latency, FAR, and computation complexity. Furthermore, the proposed adoption of soft labels more precisely demonstrates the process of seizures from ictal to postictal states. The soft labels help patients better understand the process of a seizure, relieve their stress, and facilitate caregivers or clinicians to develop personalized healthcare and clinical treatment strategies.

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